

Dhananjay Balakrishnan

University Email: dhananjb@stanford.edu | Personal Webpage: <https://dhananjay42.github.io/>

EDUCATION

Stanford University

Expected June 2026

MS in Computational and Mathematical Engineering - GPA: 3.88/4.00

Stanford, CA, USA

- Current Coursework: *Numerical Linear Algebra, Theory of Statistics, PDEs in Applied Math, Applied Causal Inference with ML & AI, Geometric and Topological analysis of Data, Introduction to Parallel Computing, Statistical Consulting Workshop*

Indian Institute of Technology, Madras

July 2019 - May 2024

Inter-disciplinary Dual Degree in Mechanical Engg. and Data Science - CGPA: 9.17/10.0

Chennai, India

- Minor in **Mathematics**
- Thesis: A Self-Supervised Approach to Motion Deblurring. Guided by [Dr. AN Rajagopalan](#). [\[Report Link\]](#)
- Awarded the IIT Madras **Governor's Gold Medal**, for **best outgoing dual-degree student**.
- Received the IIT Madras **Institute Blues Certificate of Merit**, given to outstanding graduating students.
- Relevant Coursework: *Pattern Recognition & Machine Learning, Reinforcement Learning, Multi-Armed Bandits, Modern Computer Vision, Big Data Laboratory, Fundamentals of Deep Learning, Numerical Optimization, Game Theory*

EXPERIENCE

Junior Machine Learning Engineer, YUX Cultural AI Lab

June 2025 - August 2025

Stanford SEED Internship [\[Work Portfolio\]](#)

Dakar, Senegal

- Advising on **AI localization efforts** for **indigenous, African-centered contexts** at YUX, a Senegal-based research and design firm, with a focus on technical strategy and cross-project support.
- Projects include **context-specific speech transcription** models for **low-resource languages** such as Wolof, **African stereotype detection** in AI systems, and **user studies** on how African communities interact with AI technologies.

Student Researcher, World Bank Health Group

March 2025 - June 2025

CME291: ICME Xplore, World Bank [\[Report Link\]](#)

Stanford, CA, USA

- Worked closely in a team of 3 to develop a **geospatial graph-based model** integrating clinic locations, road network data, and internet speeds to **assess healthcare accessibility** and **identify underserved regions** in South Africa.

Commodities Strats and Quant Intern, Fixed Income Division

May 2023 - August 2023

Morgan Stanley

Mumbai, India

- Implemented various mathematical models associated with volatility & risk, option pricing, and Greek calculation in C++.
- Helped migrate from the **JWSS7** to the **CiQuick** model to better fit the **volatility smiles** data, leading to an improvement of $\sim 7\%$ in the explained PnL (Profit & Loss). This has been **pushed into production**.

Project Associate, Crack Detection and Segmentation

June 2021 - October 2022

Computer Vision Club, IIT Madras

Chennai, India

- Compiled a uniformly annotated **open-source dataset** with **9000+ image-mask pairs** for the crack segmentation task.
- Used **ViT-generated features** to improve the generalizability of the model, further improving the MIOU by $\sim 5\%$.
- Co-Author: "CrackSeg9k: A Collection and Benchmark for Crack Segmentation Datasets and Frameworks", CVCIE Workshop, ECCV 2022, Tel Aviv. [\[Full-Text\]](#) [\[arXiv\]](#)

ACADEMIC PROJECTS & COMPETITIONS

Structural Discrimination in Indian AI Systems

March 2025

Guide: [Anka Reuel](#). Course: CS134 - Introduction to AI Governance. [\[Report Link\]](#)

- Used textual embeddings to demonstrate the tendency of state-of-the-art multilingual language models to show caste bias, measured across various Indian regional languages.
- Brought out potential caste bias that could arise from using an LLM as a hiring agent using a simulated cover letter dataset

Indic Language Transliteration using Seq2Seq Models

May 2023

Guide: [Dr Mitesh Khapra](#). Course: CS6910 - Fundamentals of Deep Learning.

- Built an **English-Tamil transliteration** model on the **Aksharantar** Dataset, with over **50k datapoints**.
- Achieved a validation accuracy of **0.4177**; brought it up to **0.4751** by adding an **attention layer** into the **RNN model**.
- Used the evolution of model predictions to make **qualitative observations** on the **semantics** of the Tamil script.

Bronze Medal, Super Resolution of Lunar Surface Images

January 2023

Inter-IIT Tech Meet 2023

Kanpur, India

- Achieved a PSNR of 22 by using an SRGAN model to perform a 16x super-resolution task on lunar surface images, provided by ISRO. Based on the performance of our model, we were awarded the **bronze medal**, from over **25 teams**.

EXTRA-CURRICULAR ACTIVITIES

- Quizzing:** Active member of the **IIT Madras Quiz Contingent** from 2019-24, part of the **winning contingent** at **Nihilanth** 2020 & 2023. Part of the quiz contingent that won the overall quiz cup at the Inter-IIT Cultural Meet 2023 & 2024. Kochi Regional Champions from over 600 teams in the TCS IT Wiz 2017, went on to finish 5th in the Nationals.
- Student Mentorship:** Mentored underprivileged students from JNV Pondicherry to prepare for the JEE Main and Advanced examinations, as part of the Avanti Fellows Pondicherry Chapter.

DEVELOPER TOOLS AND LANGUAGES

Frameworks: Python, C, C++, MATLAB, Wolfram, Julia, JavaScript, PyTorch, Tensorflow, Keras, NumPy, Sci-kit Learn, Open-CV, SciPy, Matplotlib, MS Excel, SQL, GNU Octave, LATEX, ABAQUS, AutoCAD, Simulink, Fusion 360

Languages: **English:** Native, **Tamil:** Native, **Malayalam:** Native, **Hindi:** Working Proficiency. **Work Permit:** US Citizen.